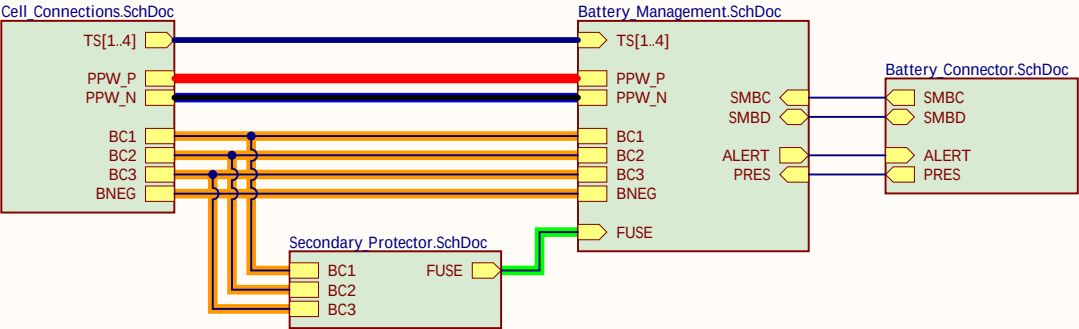
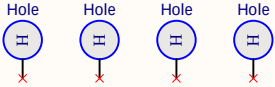


Robot Battery Pack BMS Module

3S2P, 11.1V (12.6V max.), 7000mAh



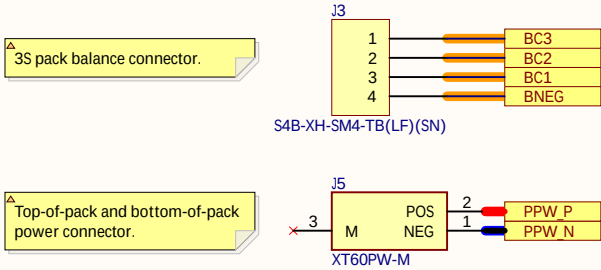
Mounting



Pack BMS Module		The ROS Robot Project		
Sub-Project: Pack_BMS_Module.PrjPcb				
Size: A4	Drawn By: Samyar Sadat Akhavi			Revision: 1.0
Date: 6/18/2026		File: Pack_BMS_Module.SchDoc		Sheet: 1 of 5

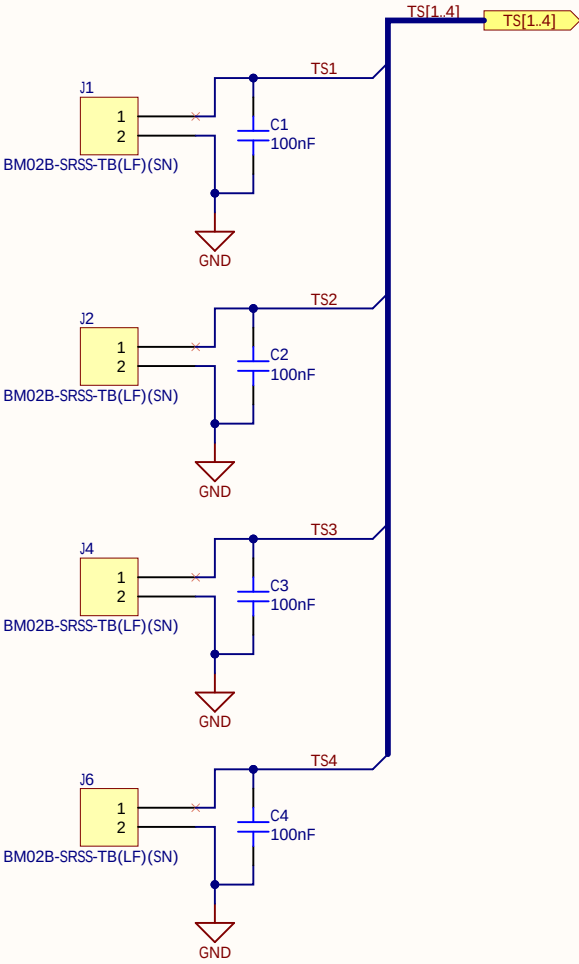
3S Pack Cells Connection

6x (3S2P) 3500mAh, Unprotected 18650 Cells



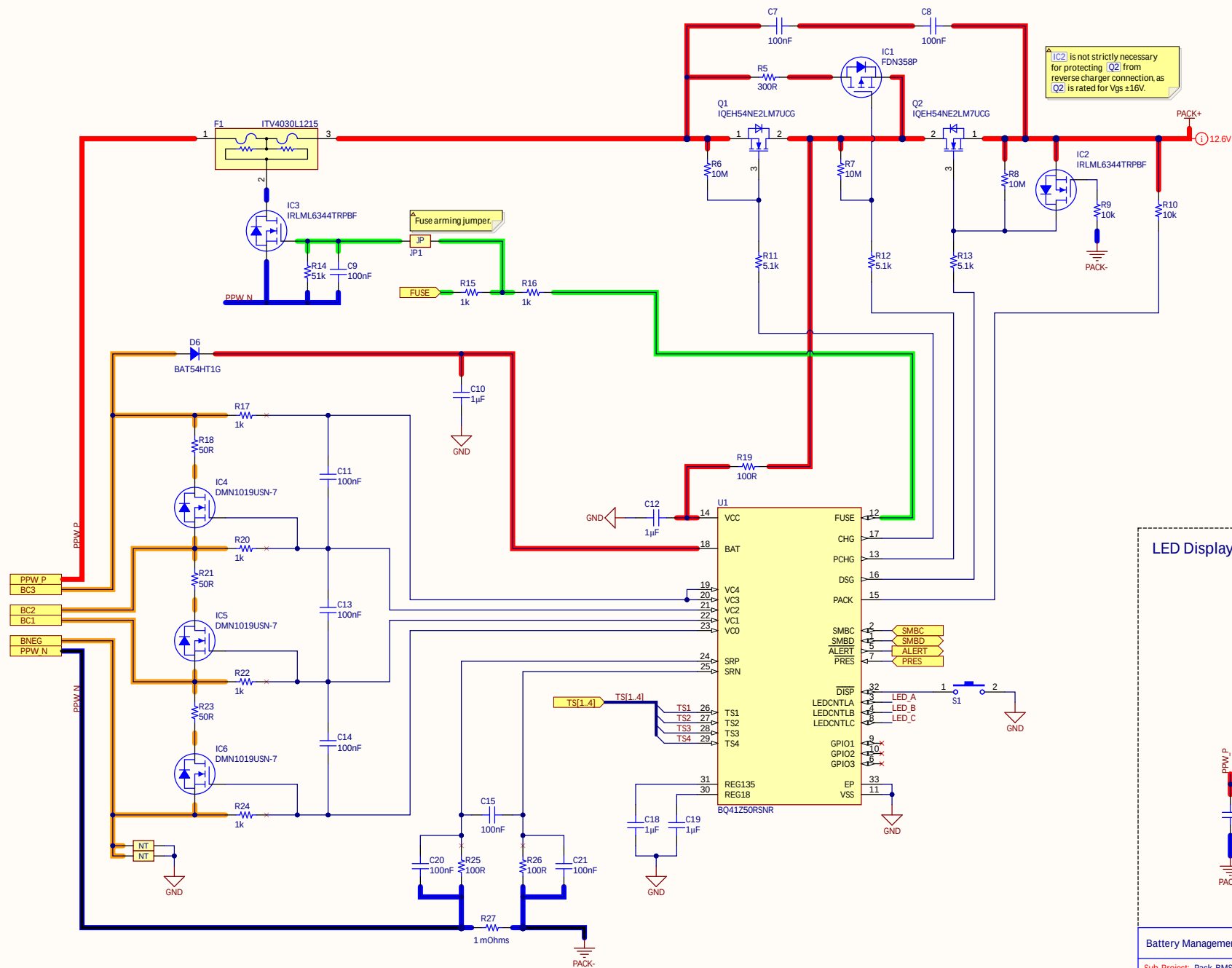
Pack Thermistors

4x 10k NTC Thermistors

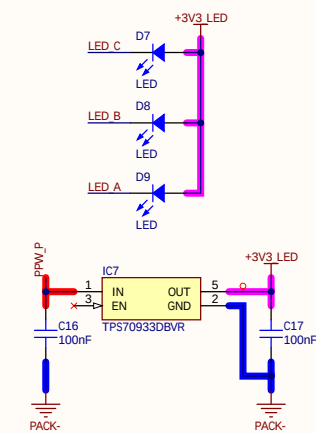


Cell Connections		The ROS Robot Project		
Sub-Project: Pack_BMS_Module.PrjPcb				
Size: A4	Drawn By: Samyar Sadat Akhavi			Revision: 1.0
Date: 6/18/2026	File: Cell_Connections.SchDoc			Sheet: 2 of 5

Battery Management & Primary Protections

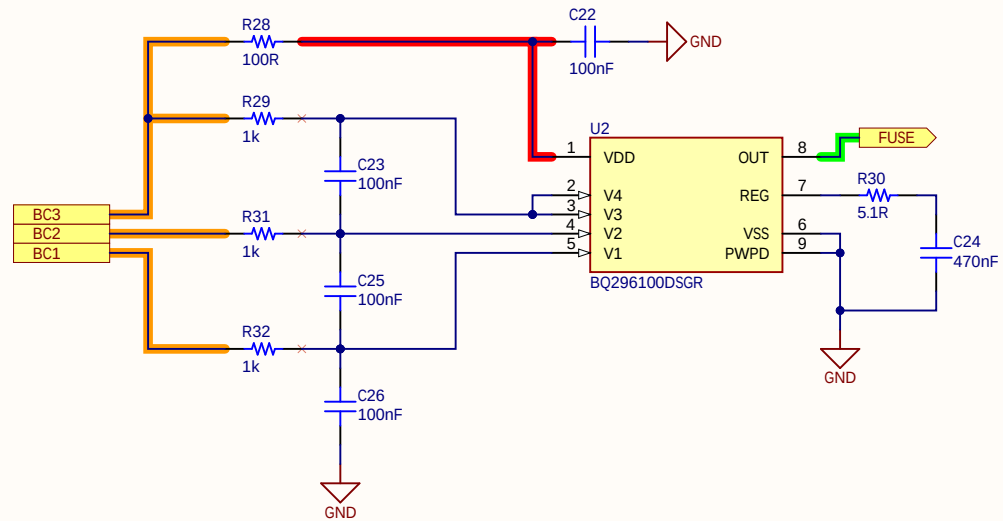


IC2 is not strictly necessary for protecting Q2 from reverse charger connection, as Q2 is rated for $V_{GS} \pm 16V$.



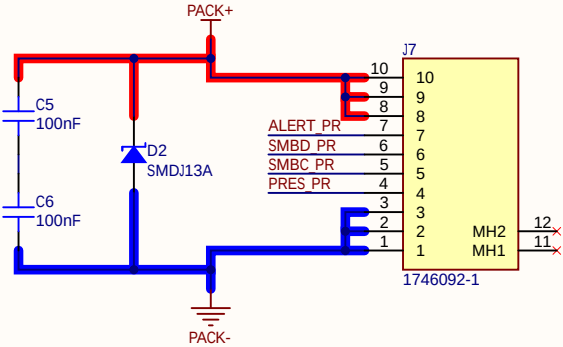
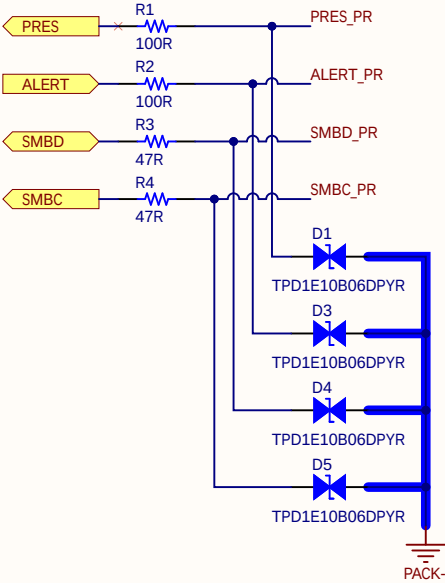
Secondary Pack Protector

Cell UV/OV Protection, Permanent Fail



Secondary Battery Protections		The ROS Robot Project		
Sub-Project: Pack_BMS_Module.PrjPcb				
Size: A4	Drawn By: Samyar Sadat Akhavi		Revision: 1.0	
Date: 6/18/2026		File: Secondary_Protector.SchDoc		Sheet: 4 of 5

Battery Pack Connector & IO ESD Protection



Battery Connector		The ROS Robot Project		
Sub-Project: Pack_BMS_Module.PrjPcb				
Size: A4	Drawn By: Samyar Sadat Akhavi			Revision: 1.0
Date: 6/18/2026	File: Battery_Connector.SchDoc			Sheet: 5 of 5

